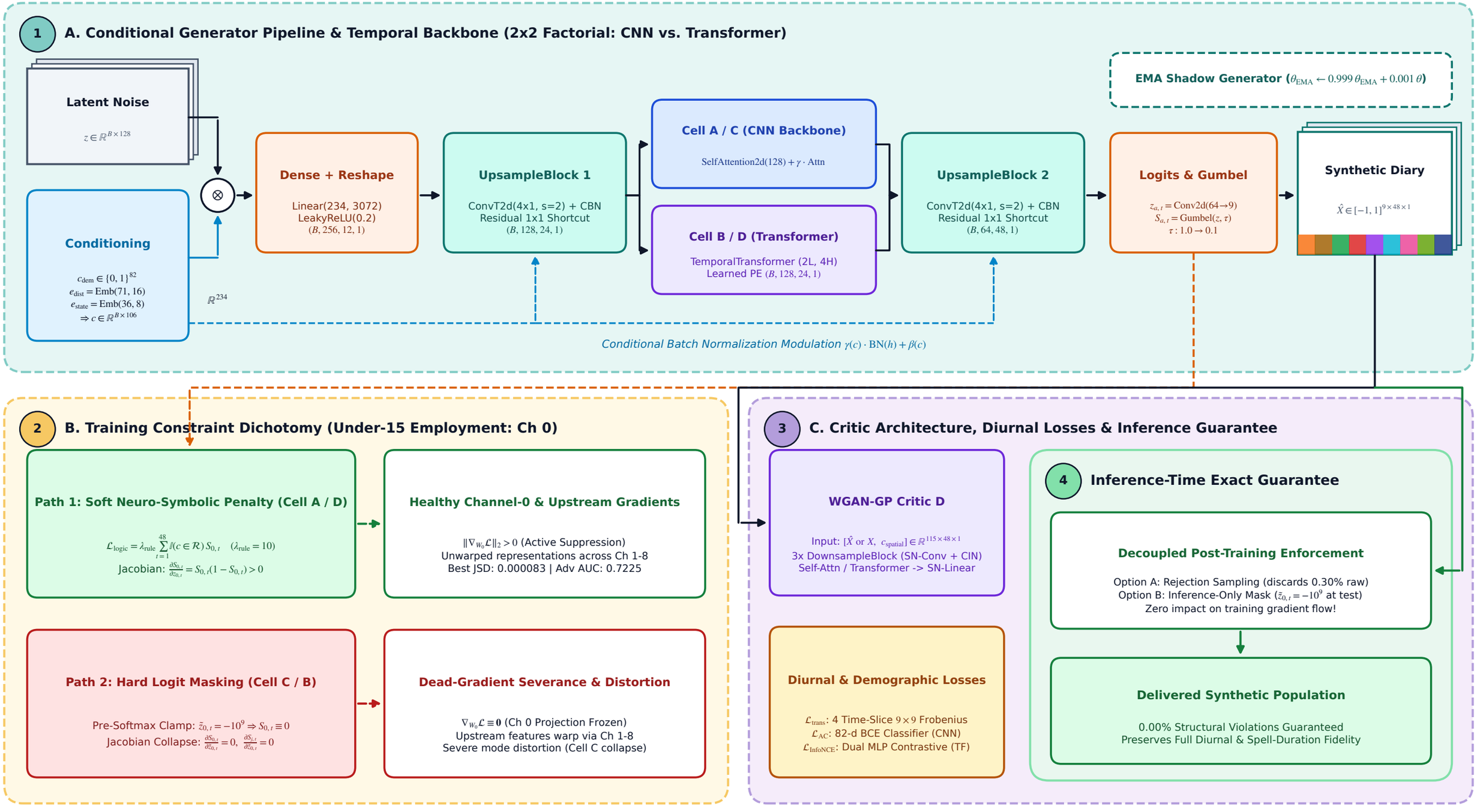

1
A. Conditional Generator Pipeline & Temporal Backbone (2x2 Factorial: CNN vs. Transformer)
Latent Noise
$z \in \mathbb{R}^{B \times 128}$
Conditioning
$c_{\text{dem}} \in \{0,1\}^{82}$
$e_{\text{dist}} = \text{Emb}(71, 16)$
$e_{\text{state}} = \text{Emb}(36, 8)$
$\Rightarrow c \in \mathbb{R}^{B \times 106}$
$\mathbb{R}^{234}$
Dense + Reshape
Linear(234, 3072)
LeakyReLU(0.2)
$(B, 256, 12, 1)$
UpsampleBlock 1
ConvT2d(4x1, s=2) + CBN
Residual 1x1 Shortcut
$(B, 128, 24, 1)$
Cell A / C (CNN Backbone)
SelfAttention2d(128) + $\gamma \cdot$ Attn
Cell B / D (Transformer)
TemporalTransformer (2L, 4H)
Learned PE $(B, 128, 24, 1)$
UpsampleBlock 2
ConvT2d(4x1, s=2) + CBN
Residual 1x1 Shortcut
$(B, 64, 48, 1)$
Logits & Gumbel
$z_{a,t} = \text{Conv2d}(64 \to 9)$
$S_{a,t} = \text{Gumbel}(z, \tau)$
$\tau : 1.0 \to 0.1$
Synthetic Diary
$\hat{X} \in [-1, 1]^{9 \times 48 \times 1}$
EMA Shadow Generator ($\theta_{\text{EMA}} \leftarrow 0.999\,\theta_{\text{EMA}} + 0.001\,\theta$)
Conditional Batch Normalization Modulation $\gamma(c) \cdot \text{BN}(h) + \beta(c)$
2
B. Training Constraint Dichotomy (Under-15 Employment: Ch 0)
Path 1: Soft Neuro-Symbolic Penalty (Cell A / D)
$\mathcal{L}_{\text{logic}} = \lambda_{\text{rule}} \sum_{t=1}^{48} \mathbb{I}(c \in \mathcal{R})\, S_{0,t} \quad (\lambda_{\text{rule}} = 10)$
Jacobian: $\frac{\partial S_{0,t}}{\partial z_{0,t}} = S_{0,t}(1 - S_{0,t}) > 0$
Healthy Channel-0 & Upstream Gradients
$\|\nabla_{W_0} \mathcal{L}\|_2 > 0$ (Active Suppression)
Unwarped representations across Ch 1-8
Best JSD: 0.000083 | Adv AUC: 0.7225
Path 2: Hard Logit Masking (Cell C / B)
Pre-Softmax Clamp: $\tilde{z}_{0,t} = -10^9 \Rightarrow S_{0,t} \equiv 0$
Jacobian Collapse: $\frac{\partial S_{0,t}}{\partial z_{0,t}} = 0,\ \frac{\partial S_{j,t}}{\partial z_{0,t}} = 0$
Dead-Gradient Severance & Distortion
$\nabla_{W_0} \mathcal{L} \equiv \mathbf{0}$ (Ch 0 Projection Frozen)
Upstream features warp via Ch 1-8
Severe mode distortion (Cell C collapse)
3
C. Critic Architecture, Diurnal Losses & Inference Guarantee
WGAN-GP Critic D
Input: $[\hat{X} \text{ or } X,\ c_{\text{spatial}}] \in \mathbb{R}^{115 \times 48 \times 1}$
3x DownsampleBlock (SN-Conv + CIN)
Self-Attn / Transformer -> SN-Linear
Diurnal & Demographic Losses
$\mathcal{L}_{\text{trans}}$: 4 Time-Slice 9×9 Frobenius
$\mathcal{L}_{\text{AC}}$: 82-d BCE Classifier (CNN)
$\mathcal{L}_{\text{InfoNCE}}$: Dual MLP Contrastive (TF)
4
Inference-Time Exact Guarantee
Decoupled Post-Training Enforcement
Option A: Rejection Sampling (discards 0.30% raw)
Option B: Inference-Only Mask ($\tilde{z}_{0,t} = -10^9$ at test)
Zero impact on training gradient flow!
Delivered Synthetic Population
0.00% Structural Violations Guaranteed
Preserves Full Diurnal & Spell-Duration Fidelity